

Optimization

- 1 Polyhedral Optimization
 - geometry, algebra, business
 - the simplex method

- 2 Unconstrained Optimization
 - local and global optima
 - gradient descent

MCS 320 Lecture 32
Introduction to Symbolic Computation
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Polyhedral Optimization

Viewed from a geometry, algebra, or business perspective:

- How far can we push a plane so it touches a polyhedron?
- Optimize a linear function subject to linear inequalities.

Two normal forms:

$$\begin{array}{ll}\text{maximize} & \mathbf{c}^T \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \leq \mathbf{b}\end{array}$$

$$\begin{array}{ll}\text{minimize} & \mathbf{c}^T \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \geq \mathbf{b}\end{array}$$

- Minimizing cost and maximizing profit are equivalent in the polyhedral world.

Optimization

1 Polyhedral Optimization

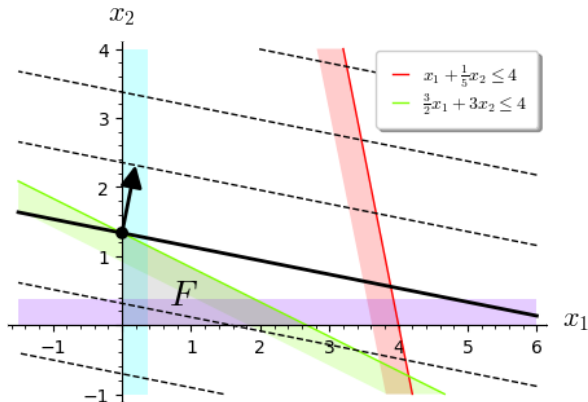
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Linear Programming with the Simplex Method

Maximize $x_1 + 5x_2$ subject to $x_1 + 0.2x_2 \leq 4$ and $1.5x_1 + 3x_2 \leq 4$.



Starting at $(x_1 = 0, x_2 = 0)$, the simplex method pivots to an adjacent corner of the feasible region that has a better optimum.

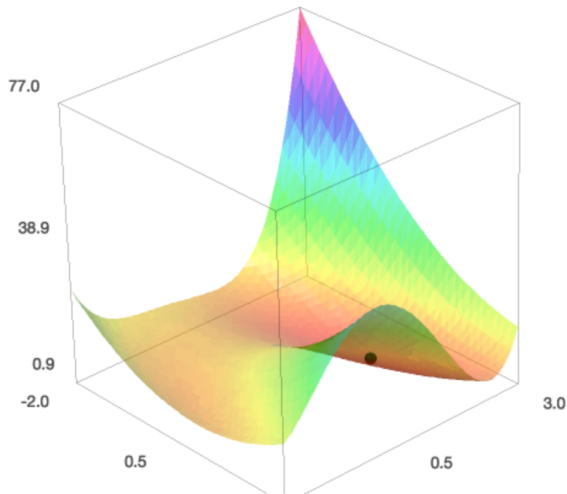
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Unconstrained Optimization

Example: minimize $z = f(x, y) = (3 + x - y^2)^2 + (x - 1)^2 + (y - 1)^2$.



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Gradient Descent

For a function f in several variables, at the current point, the gradient descent method

- 1 evaluates the gradient ∇f at the current point,
- 2 updates the current point with a step in the ∇f direction.

The gradient descent method can find only a local optimum.

To find all optima,

- 1 compute all critical points, solve $\nabla f = 0$,
- 2 classify the points by their values in f .